Newman-Penrose Formalism Calculations

Stephani et al., Equation (13.55_e1)

Metric

$$ds^2 = -\frac{\cosh\left(k\tau\right) e^{\left(2\,A\tau+\frac{k^2\tau}{2\,A}\right)}}{k}\mathrm{d}\tau\otimes\mathrm{d}\tau + \frac{k}{\cosh\left(k\tau\right)}\mathrm{d}x\otimes\mathrm{d}x - \frac{kz}{\cosh\left(k\tau\right)}\mathrm{d}x\otimes\mathrm{d}y - \frac{kz}{\cosh\left(k\tau\right)}\mathrm{d}y\otimes\mathrm{d}x + \left(\frac{k^2z^2+\cosh\left(k\tau\right)^2 e^{\left(2\,A\tau\right)}}{k\cosh\left(k\tau\right)}\right)\mathrm{d}y\otimes\mathrm{d}y + \frac{\cosh\left(k\tau\right) e^{\left(\frac{k^2\tau}{2\,A}\right)}}{k}\mathrm{d}z\otimes\mathrm{d}z$$

Coordinates

$$(x^\mu) = (\tau, x, y, z)$$

Metric Functions

$$X^2 = \frac{\cosh\left(k\tau\right)}{k}$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\sqrt{2}\sqrt{\frac{\cosh\left(k\tau\right)}{k}}e^{\left(A\tau+\frac{k^2\tau}{4\,A}\right)}\right)d\tau + \left(\frac{1}{2}\sqrt{2}\sqrt{\frac{\cosh\left(k\tau\right)}{k}}e^{\left(\frac{k^2\tau}{4\,A}\right)}\right)dz$$

$$n = \left(\frac{1}{2}\sqrt{2}\sqrt{\frac{\cosh\left(k\tau\right)}{k}}e^{\left(A\tau+\frac{k^2\tau}{4\,A}\right)}\right)d\tau + \left(-\frac{1}{2}\sqrt{2}\sqrt{\frac{\cosh\left(k\tau\right)}{k}}e^{\left(\frac{k^2\tau}{4\,A}\right)}\right)dz$$

$$m = \left(\frac{\sqrt{2}}{2\sqrt{\frac{\cosh(k\tau)}{k}}}\right)dx + \left(\frac{1}{2}i\sqrt{2}\sqrt{\frac{\cosh\left(k\tau\right)}{k}}e^{\left(A\tau\right)} - \frac{\sqrt{2}z}{2\sqrt{\frac{\cosh(k\tau)}{k}}}\right)dy$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2\sqrt{\frac{\cosh(k\tau)}{k}}} \right) dx + \left(-\frac{1}{2} i\sqrt{2}\sqrt{\frac{\cosh\left(k\tau\right)}{k}} e^{\left(A\tau\right)} - \frac{\sqrt{2}z}{2\sqrt{\frac{\cosh(k\tau)}{k}}} \right) dy$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}\sqrt{k}e^{\left(-A\tau - \frac{k^2\tau}{4A}\right)}}{2\sqrt{\cosh\left(k\tau\right)}} \right) d\tau + \left(\frac{\sqrt{2}\sqrt{k}e^{\left(-\frac{k^2\tau}{4A}\right)}}{2\sqrt{\cosh\left(k\tau\right)}} \right) dz$$

$$n = \left(-\frac{\sqrt{2}\sqrt{k}e^{\left(-A\tau - \frac{k^2\tau}{4A}\right)}}{2\sqrt{\cosh\left(k\tau\right)}} \right) d\tau + \left(-\frac{\sqrt{2}\sqrt{k}e^{\left(-\frac{k^2\tau}{4A}\right)}}{2\sqrt{\cosh\left(k\tau\right)}} \right) dz$$

$$m = \left(\frac{i\sqrt{2}\sqrt{k}ze^{\left(-A\tau\right)}}{2\sqrt{\cosh\left(k\tau\right)}} + \frac{\sqrt{2}\sqrt{\cosh\left(k\tau\right)}}{2\sqrt{k}} \right) dx + \left(\frac{i\sqrt{2}\sqrt{k}e^{\left(-A\tau\right)}}{2\sqrt{\cosh\left(k\tau\right)}} \right) dy$$

$$\overline{m} = \left(-\frac{i\sqrt{2}\sqrt{k}ze^{\left(-A\tau\right)}}{2\sqrt{\cosh\left(k\tau\right)}} + \frac{\sqrt{2}\sqrt{\cosh\left(k\tau\right)}}{2\sqrt{k}} \right) dx + \left(-\frac{i\sqrt{2}\sqrt{k}e^{\left(-A\tau\right)}}{2\sqrt{\cosh\left(k\tau\right)}} \right) dy$$

Directional Derivatives

$$D = -\frac{\sqrt{2}\sqrt{k}e^{\left(-A\tau - \frac{k^2\tau}{4A}\right)}}{2\sqrt{\cosh\left(k\tau\right)}} \partial_\tau + \frac{\sqrt{2}\sqrt{k}e^{\left(-\frac{k^2\tau}{4A}\right)}}{2\sqrt{\cosh\left(k\tau\right)}} \partial_z$$

$$\Delta = -\frac{\sqrt{2}\sqrt{k}e^{\left(-A\tau - \frac{k^2\tau}{4A}\right)}}{2\sqrt{\cosh\left(k\tau\right)}} \partial_\tau - \frac{\sqrt{2}\sqrt{k}e^{\left(-\frac{k^2\tau}{4A}\right)}}{2\sqrt{\cosh\left(k\tau\right)}} \partial_z$$

$$\delta = \frac{i\sqrt{2}\sqrt{k}ze^{\left(-A\tau\right)}}{2\sqrt{\cosh\left(k\tau\right)}} + \frac{\sqrt{2}\sqrt{\cosh\left(k\tau\right)}}{2\sqrt{k}} \partial_x + \frac{i\sqrt{2}\sqrt{k}e^{\left(-A\tau\right)}}{2\sqrt{\cosh\left(k\tau\right)}} \partial_y$$

$$\overline{\delta} = -\frac{i\sqrt{2}\sqrt{k}ze^{(-A\tau)}}{2\sqrt{\cosh(k\tau)}} + \frac{\sqrt{2}\sqrt{\cosh(k\tau)}}{2\sqrt{k}}\partial_x - \frac{i\sqrt{2}\sqrt{k}e^{(-A\tau)}}{2\sqrt{\cosh(k\tau)}}\partial_y$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}A\sqrt{k}e^{\left(-A\tau - \frac{k^2\tau}{4A}\right)}}{4\sqrt{\cosh(k\tau)}}$$

$$\sigma = -\frac{\sqrt{2}A\sqrt{k}e^{\left(-A\tau - \frac{k^2\tau}{4A}\right)}}{4\sqrt{\cosh(k\tau)}} - \frac{\sqrt{2}k^{\frac{3}{2}}e^{\left(-A\tau - \frac{k^2\tau}{4A}\right)}\sinh(k\tau)}{4\cosh(k\tau)^{\frac{3}{2}}} + \frac{i\sqrt{2}k^{\frac{3}{2}}e^{\left(-A\tau - \frac{k^2\tau}{4A}\right)}}{4\cosh(k\tau)^{\frac{3}{2}}}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}A\sqrt{k}e^{\left(-A\tau - \frac{k^2\tau}{4A}\right)}}{4\sqrt{\cosh(k\tau)}}$$

$$\lambda = \frac{\sqrt{2}A\sqrt{k}e^{\left(-A\tau - \frac{k^2\tau}{4A}\right)}}{4\sqrt{\cosh(k\tau)}} + \frac{\sqrt{2}k^{\frac{3}{2}}e^{\left(-A\tau - \frac{k^2\tau}{4A}\right)}\sinh(k\tau)}{4\cosh(k\tau)^{\frac{3}{2}}} - \frac{i\sqrt{2}k^{\frac{3}{2}}e^{\left(-A\tau - \frac{k^2\tau}{4A}\right)}}{4\cosh(k\tau)^{\frac{3}{2}}}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2}k^{\frac{5}{2}}e^{\left(-A\tau - \frac{k^2\tau}{4\,A}\right)}}{16\,A\sqrt{\cosh\left(k\tau\right)}} - \frac{\sqrt{2}k^{\frac{3}{2}}e^{\left(-A\tau - \frac{k^2\tau}{4\,A}\right)}\sinh\left(k\tau\right)}{8\,\cosh\left(k\tau\right)^{\frac{3}{2}}} + \frac{i\,\sqrt{2}k^{\frac{3}{2}}e^{\left(-A\tau - \frac{k^2\tau}{4\,A}\right)}}{8\,\cosh\left(k\tau\right)^{\frac{3}{2}}}$$

$$\gamma = \frac{\sqrt{2}k^{\frac{5}{2}}e^{\left(-A\tau - \frac{k^2\tau}{4\,A}\right)}}{16\,A\sqrt{\cosh\left(k\tau\right)}} + \frac{\sqrt{2}k^{\frac{3}{2}}e^{\left(-A\tau - \frac{k^2\tau}{4\,A}\right)}\sinh\left(k\tau\right)}{8\,\cosh\left(k\tau\right)^{\frac{3}{2}}} - \frac{i\,\sqrt{2}k^{\frac{3}{2}}e^{\left(-A\tau - \frac{k^2\tau}{4\,A}\right)}}{8\,\cosh\left(k\tau\right)^{\frac{3}{2}}}$$

Ricci Tensor Components

$\Phi_{00} = 0$

$\Phi_{01} = 0$

$\Phi_{02} = 0$

$\Phi_{10} = 0$

$\Phi_{11} = 0$

$\Phi_{12} = 0$

$\Phi_{20} = 0$

$\Phi_{21} = 0$

$\Phi_{22} = 0$

$\Lambda = 0$

Weyl Tensor Components

$$\Psi_0 = -\frac{3\,k^3 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}}{8\,\cosh\left(k\tau\right)} - \frac{Ak^2 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}\sinh\left(k\tau\right)}{4\,\cosh\left(k\tau\right)^2} - \frac{k^4 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}\sinh\left(k\tau\right)}{8\,A\cosh\left(k\tau\right)^2} + \frac{i\,Ak^2 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}}{4\,\cosh\left(k\tau\right)^2} + \frac{i\,k^4 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}}{8\,A\cosh\left(k\tau\right)^2} + \frac{3i\,k^3 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}\sinh\left(k\tau\right)}{4\,\cosh\left(k\tau\right)^3} + \frac{3\,k^3 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}}{4\,\cosh\left(k\tau\right)^3}$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{Ak^2 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}\sinh\left(k\tau\right)}{4\,\cosh\left(k\tau\right)^2} + \frac{k^3 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}\sinh\left(k\tau\right)^2}{8\,\cosh\left(k\tau\right)^3} - \frac{i\,Ak^2 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}}{4\,\cosh\left(k\tau\right)^2} - \frac{i\,k^3 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}\sinh\left(k\tau\right)}{4\,\cosh\left(k\tau\right)^3} - \frac{k^3 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}}{8\,\cosh\left(k\tau\right)^3}$$

$$\Psi_3 = 0$$

$$\Psi_4 = -\frac{3\,k^3 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}}{8\,\cosh\left(k\tau\right)} - \frac{Ak^2 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}\sinh\left(k\tau\right)}{4\,\cosh\left(k\tau\right)^2} - \frac{k^4 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}\sinh\left(k\tau\right)}{8\,A\cosh\left(k\tau\right)^2} + \frac{i\,Ak^2 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}}{4\,\cosh\left(k\tau\right)^2} + \frac{i\,k^4 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}}{8\,A\cosh\left(k\tau\right)^2} + \frac{3i\,k^3 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}\sinh\left(k\tau\right)}{4\,\cosh\left(k\tau\right)^3} + \frac{3\,k^3 e^{\left(-2\,A\tau - \frac{k^2\tau}{2\,A}\right)}}{4\,\cosh\left(k\tau\right)^3}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"